

# Leong Ko Ryan Jasper

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## EDUCATION

### National University of Singapore

Singapore

*Master of Science in Computer Engineering, Specialisation in Machine Intelligence Applications*

*Aug 2026 – Jun 2027*

*Bachelor of Engineering (First Class Honours), Computer Engineering*

*Aug 2023 – Aug 2026*

- GPA: 4.75/5.0 | Dean's List in all eligible semesters (2/2) | Engineering Scholars Programme

### Stanford University

Stanford, CA

*Visiting Student, Spring Quarter*

*Apr 2025 – Jun 2025*

## EXPERIENCE

### Robotics Software Engineer Intern

May 2026 – Present

*DSO National Laboratories*

*Singapore*

- Built an agentic robot task-planning harness inspired by SayPlan and LookPlanGraph, integrating local LLM tool use, DDS middleware, and hierarchical 3D scene graphs for spatial reasoning and closed-loop replanning
- Developed low-fidelity robot embodiment simulators and DDS adapters for external simulators, enabling hardware-independent testing, scenario replay, telemetry, and end-to-end autonomy validation
- Benchmarked 40+ compact language models for task success, tool-use reliability, inference latency, and resource usage, producing a reproducible evaluation pipeline for local and edge deployment

### Robotics Software Engineer

Jan 2025 – Dec 2025

*ZeroshotData — Early-Stage Robotics Startup*

*San Francisco, CA*

- Built robotics infrastructure on GCP using Terraform and Kubernetes, deploying 20+ containerized services with CI/CD and FastAPI-based device registration, telemetry, monitoring, and fleet operations
- Integrated visual-inertial ORB-SLAM3 into a UMI gripper using OAK-D stereo cameras, Kalibr camera-IMU calibration, Allan variance analysis, and synchronized sensor acquisition
- Built Python applications for the end-to-end processing pipeline behind the [SF Fold dataset](#) across 100+ hours of robot-folding demonstrations, including video preprocessing and synchronization, automated data validation, ORB-SLAM3 execution, generation of review videos and Rerun visualizations, and Mosaic-based dataset sharding
- Automated robot and sensor calibration using a Piper 6-DoF arm and prototyped infrared tracking with Arducam cameras, hardware frame synchronization, and Orange Pi compute
- Engineered an electromagnetic full-body tracking system with a 1.5 m range, centimetre-level positional accuracy, and 5° rotational accuracy using custom PCBs, STM32 firmware, Helmholtz coils, and Onshape CAD

### Research Engineering Intern

May 2024 – Aug 2024

*A\*STAR — National Metrology Centre*

*Singapore*

- Built a Python measurement platform using PyVISA, Pandas, and Dash on an Orange Pi, and designed a Verilog FPGA frequency synthesizer achieving 6.5–9.0 ns timing precision

### Software Engineering Intern

Apr 2023 – Jun 2023

*HolyWally — Fintech Startup*

*Singapore*

- Built Python and SQL tools backed by PostgreSQL for an operational dashboard used by approximately 100 employees

## PROJECTS

[Flair AI](#) | *React, Gemini, Twilio*

2025

- **Winner, 1st Place (\$1,500) — AI for Good Hackathon.** Built an AI writing assistant with automated grading and a Gemini/Twilio voice-feedback agent for approximately 400 students

## SKILLS

|                                   |                                                                                                                                                                           |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming &amp; ML:</b>      | Python, C++, TypeScript/JavaScript, SQL, Bash; PyTorch, NumPy, Pandas, OpenCV, llama.cpp, FastAPI, React, pytest                                                          |
| <b>Robotics &amp; Autonomy:</b>   | ROS 2, DDS, Gazebo, ORB-SLAM3, visual-inertial odometry (VIO), sensor fusion, camera-IMU calibration, Kalibr, DepthAI, OAK-D, 3D scene graphs, task planning, URDF, Rerun |
| <b>Systems &amp; Cloud:</b>       | Linux, Git, REST APIs, GCP, Kubernetes (K8s), Terraform, Docker, CI/CD, PostgreSQL                                                                                        |
| <b>Embedded &amp; Interfaces:</b> | STM32/RaspberryPi/OrangePi, Verilog/FPGA, CAN, MIPI CSI-2, USB, UART, I <sup>2</sup> C, SPI, MQTT                                                                         |
| <b>Hardware &amp; CAD:</b>        | KiCad, PCB design and bring-up, soldering, Onshape CAD, 3D printing, oscilloscopes, electromechanical prototyping, sensor bring-up                                        |
| <b>Languages:</b>                 | English, Mandarin Chinese (Limited Working Proficiency)                                                                                                                   |